## 7.6.7 The eye

Eye is a natural optical instrument human beings have. As the eye lens is flexible, its focal length can be changed to some extent. When the eye is fully relaxed, its focal length is maximum and when it is strained its focal length is minimum. The image must be formed on the retina for clear vision. The diameter of eye ball for a normal adult is about 2.5 cm. Hence, the distance between eye lens and retina (image distance) is fixed always at 2.5 cm. We can just discuss the optical functioning of eye without giving importance to the refractive indices of the two liquids, aqueous humor and virtuous humor present in the eye. A person with normal vision can see objects kept at infinity in the relaxed condition with a maximum focal length $f_{max}$ of the eye lens as shown in Figure 7.46(a) and in the strained condition, with a minimum focal length $f_{min}$ for an object kept at near point $D$ (25 cm) as shown in Figure 7.46(b).

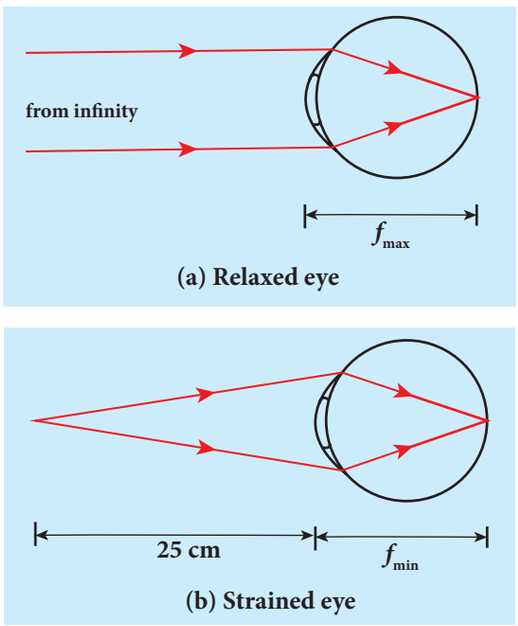



**Figure 7.46** Focusing of normal eye

Let us find $f_{max}$ and $f_{min}$ of human eye from the lens equation (6.63).

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

When the object is at infinity, $u = -\infty$, and $v$ = 2.5 cm (distance between eye lens and retina), the eye can see the object in relaxed condition with $f_{max}$. Substituting these values in the lens equation gives,

$$\frac{1}{f_{\max}}=\frac{1}{2.5\ \text{cm}}-\frac{1}{-\infty}$$

$$f_{\max}=2.5\ \text{cm}$$

When the object is at near point, $u$ = –25 cm and $v$ = 2.5 cm, the eye can see the object in strained condition with $f_{min}$. Substituting these values in the lens equation gives,

$$\frac{1}{f_{\min}}=\frac{1}{2.5\ \text{cm}}-\frac{1}{-25\ \text{cm}}$$

$$f_{\min}=2.27\ \text{cm}$$

This implies that by varying the focal length of the eye lens by a small value of $f_{max} - f_{min}$ = 0.23 cm, a person can see the objects from infinity to the near point. Now, we can discuss some common defects of vision in the eye.

### 7.6.7.1 Nearsightedness (*myopia*)

A person suffering from *nearsightedness* (or) *myopia* cannot see distant objects clearly. This may be due to the short focal length of the eye lens (or) larger diameter of the eyeball than usual. These people have difficulty in relaxing their eye to the extent of what is needed. Thus, they need correcting lens.

For them, parallel rays coming from the distant object get focused before reaching the retina as shown in Figure 7.47(a). But, these persons can see objects which are nearer. Let $x$ be the maximum distance up to which a person with nearsightedness



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